



UNIVERSIDADE D
COIMBRA

Movie Recommendation System SRS - ASE

Bernardo Miguel Gaspar de Sousa Coelho Lopes, n^o2020222465

David Miguel Costa da Silva, n^o2020217642

Marcelo Rodrigues Gomes, n^o2021222994

João Pedro Gomes Laranjeiro, n^o2024181464

Simão Tomás Botas Carvalho, n^o2021223055

October 2025

Contents

1	Introduction	3
1.1	Purpose	3
1.2	Scope	3
1.3	Definitions, Acronyms, Abbreviations	3
1.4	Technologies	4
2	System Overview	5
2.1	System Description	5
2.2	Key Features	5
2.3	Stakeholders	5
3	Requirements	6
3.1	User Stories	6
3.2	Functional Requirements	7
3.3	Functional Requirements - Use Cases	9
3.4	Non-Functional Requirements	19
4	System Design	22
4.1	C4 System Context Diagram	22
4.2	Architecture Overview	23
4.3	Database Design (Conceptual)	23
5	Assumptions and Constraints	25
5.1	Assumptions	25
5.2	Constraints	26

1 Introduction

1.1 Purpose

The purpose of this project is to build and implement a movie recommendation system that bases its recommendations on the current user's past information. A few key points it can use will be: movie genre, director, rating, actors, era, time length. We will use two approach methods to building this project: the traditional way of DevOps approach and AI-assisted coding/building. The main purpose of this project is to show how the correct use of AI can help coding and improve workflow.

1.2 Scope

The system being developed is a movie recommendation platform designed to help users discover new titles that match their interests. Its boundaries include a complete yet manageable ecosystem composed of a backend service, a frontend interface, and a database to store users, movies, and ratings. The system will focus on providing a functional, realistic prototype suitable for academic and learning purposes.

The main features within the scope of this project include user management, allowing people to register, log in, and maintain personal profiles. Users will be able to browse a catalog of movies, search for titles by keywords, genre, author, or director, and access detailed information about each item. A rating system will let users evaluate movies, helping the system refine its recommendations towards that particular user. The recommendation engine will then use these ratings, along with user preferences, to suggest new items that align with individual tastes.

1.3 Definitions, Acronyms, Abbreviations

This section defines the main technical terms and abbreviations used throughout the document to ensure clarity and consistency.

- **CI/CD (Continuous Integration and Continuous Deployment):** Practices that automate code integration, testing, and deployment, ensuring faster and more reliable releases.
- **LLM (Large Language Model):** An AI model capable of understanding and generating natural language, used here to assist in coding, documentation, and testing.
- **API (Application Programming Interface):** A communication bridge between the frontend and backend that allows data exchange and functionality integration.
- **SDLC (Software Development Life Cycle):** The process that covers all stages of software creation, from requirements to deployment.
- **DevOps:** A set of practices that combines development and operations to improve automation, collaboration, and delivery speed.

- **UI/UX (User Interface / User Experience):** The design and usability aspects that determine how users interact with and perceive the system.
- **DBMS (Database Management System):** Software used to store, organize, and manage the system's data efficiently.
- **MVP (Minimum Viable Product):** The simplest version of a product that includes only the essential features needed to validate core functionality and gather user feedback for future improvements.

1.4 Technologies

The system will be developed using the following technologies:

- **FastAPI** – for building the backend RESTful API.
- **SQLAlchemy** – as the Object-Relational Mapper (ORM) for database interactions.
- **PostgreSQL** – as the primary relational database.
- **React** – for developing the frontend user interface.
- **GitLab** – for version control, testing, and CI/CD deployment.

2 System Overview

2.1 System Description

The system is a movie recommendation platform that provides personalized suggestions based on user preferences and interactions. Users can browse a catalog, search for items, rate what they've consumed, and receive tailored recommendations. Technically, it consists of three layers: a frontend for user interaction, a backend for processing requests and business logic, and a database for storing users, items, and ratings. A CI/CD pipeline ensures automated testing and deployment for consistent performance.

2.2 Key Features

Key features include search by keywords, genres, or creators; catalog browsing with detailed item information; and user profile management. The rating system influences recommendations, which are generated using collaborative or content-based filtering.

2.3 Stakeholders

Primary stakeholders are end users, whose interactions drive the recommendation engine. The development team handles design, coding, testing, and deployment, while project managers coordinate planning and timelines. AI tools, such as LLMs, assist in code generation, testing, and documentation, supporting efficient development and system functionality in the version of the project that will use such tools.

3 Requirements

3.1 User Stories

Traditional

- Story1: As a new user, I want to register my account so that I can use the system.
- Story2: As a user, I want to log in so that I can access my account.
- Story3: As a user, I want to be able to access my profile so that I can manage it.
- Story4: As a user, I want to be able to view a list of movies so that I can discover new ones.
- Story5: As a user, I want to be able to search movies so that I can see their description.
- Story6: As a user, I want to be able to rate movies so that other users can see the general rating.
- Story7: As a user, I want to be recommended movies based on my interests so that I can discover new movies.

Prompt used

Context was given. Prompt: Review these user stories that i have already defined. Suggest additional user stories.

AI-assisted

Authentication

1. As a new user, I want to register my account so that I can use the system.
2. As a user, I want to log in so that I can access my account.
3. As a user, I want to log out so that I can secure my session on shared devices.

Profile Management

1. As a user, I want to view my profile so that I can see my account information and rating history.
2. As a user, I want to edit my profile information so that I can keep my account details current.

Browsing & Discovery

1. As a user, I want to view a list of movies so that I can discover new content.
2. As a user, I want to view detailed information about a movie so that I can decide if I want to watch it.
3. As a user, I want to search for movies by title, genre, or director so that I can quickly find specific movies.

4. As a user, I want to filter movies by genre so that I can browse categories I enjoy.

Ratings & Recommendations

1. As a user, I want to rate movies on a scale of 1-5 stars so that I can receive personalized recommendations.
2. As a user, I want to view the average rating and number of ratings for a movie so that I can gauge its popularity.
3. As a user, I want to receive personalized movie recommendations based on my rating history so that I can discover similar movies.

3.2 Functional Requirements

Traditional

User Authentication

FR1.1: The system shall allow new users to register with email and password.

FR1.2: The system shall allow registered users to log in with their credentials.

FR1.3: The system shall allow users to log out of their session.

User Profile Management

FR2.1: The system shall display user profile information including username and rating history.

Movie Browsing

FR3.1: The system shall display a list of available movies.

FR3.2: The system shall display detailed movie information including title, genre, director, description, average rating, and total ratings.

FR3.3: The system shall allow users to search movies by title.

FR3.4: The system shall allow users to filter movies by genre.

Movie Ratings

FR4.1: The system shall allow users to rate movies on a 1–5 star scale.

FR4.2: The system shall calculate and display the average rating for each movie.

Personalized Recommendations

FR5.1: The system shall generate personalized movie recommendations based on a user's rating history.

FR5.2: The system shall display recommended movies to users on a dedicated recommendations page.

AI-Assisted

User Authentication

- FR1.1:** The system shall allow new users to register by providing email and password.
- FR1.2:** The system shall reject registration if the password is less than 8 characters.
- FR1.3:** The system shall allow registered users to log in by providing email and password.
- FR1.4:** The system shall create a session token upon successful login.
- FR1.5:** The system shall display an error message “Invalid email or password” when login credentials do not match any registered user.
- FR1.6:** The system shall allow authenticated users to log out and terminate their session.

User Profile

- FR2.1:** The system shall display the user profile page showing username and email address.
- FR2.2:** The system shall allow users to update their username.
- FR2.3:** The system shall allow users to update their email address.
- FR2.4:** The system shall allow users to change their password by providing their current password and a new password (minimum 8 characters).
- FR2.5:** The system shall display a list of all movies the user has rated, showing movie title and star rating (1–5).

Movie Browsing

- FR3.1:** The system shall display a list of all available movies with title, genre, and average rating visible.
- FR3.2:** The system shall display detailed movie information showing title, genre, director, description, average rating (rounded to one decimal place), and total number of ratings.
- FR3.3:** The system shall allow users to search for movies by entering a title search term.
- FR3.4:** The system shall return all movies where the title contains the search term (case-insensitive, partial match).
- FR3.5:** The system shall allow users to filter the movie list by selecting a genre.
- FR3.6:** The system shall display only movies matching the selected genre when a filter is applied.

Movie Ratings

- FR4.1:** The system shall allow authenticated users to rate any movie by selecting 1, 2, 3, 4, or 5 stars.
- FR4.2:** The system shall calculate the average rating as the arithmetic mean of all user ratings for a movie, rounded to one decimal place.
- FR4.3:** The system shall display the total count of user ratings for each movie as an integer.

FR4.4: The system shall allow users to update their existing rating for a movie by submitting a new rating value.

FR4.5: The system shall prevent unauthenticated users from submitting ratings.

Recommendations

FR5.1: The system shall generate up to 10 movie recommendations for users who have submitted at least one rating.

FR5.2: The system shall recommend movies from genres where the user has given ratings of 4 or 5 stars.

FR5.3: The system shall exclude movies the user has already rated from the recommendations list.

FR5.4: The system shall display the 10 movies with the highest average rating (minimum of three total ratings) for users who have not yet rated any movies.

FR5.5: The system shall display each recommended movie with title, genre, and average rating.

3.3 Functional Requirements - Use Cases

Traditional

Catalog Management

Table 1: Use Case 001 - Browse Movies

Use Case 001 - Browse Movies	
Description	Allows the user to browse movies presented in the catalog.
Scope	Movies Catalog
Level	User Level
Preconditions	User has access to the movies catalog page.
Trigger	User opens the catalog page
Minimal Guarantees	No movie is deleted.
Main Success Scenario	
User	1. Access the movie catalog
System	2. Displays the list of movies with title, genre, release year, and cover.
User	3. User navigates through the list 4. The user selects the movie
System	5. Shows a detailed page of the selected movie
Exceptions	
2a - The system does not have the movies	
• 2a1. The system presents the message “No movies are available yet.”	
Post-conditions	The user can see the available movies

Table 2: Use Case 002 - View Movies Details

Use Case 002 - View Movies Details	
Description	Allows the user to view details of the selected movie.
Scope	Movies Catalog
Level	User Level
Preconditions	• User has access to the movies catalog page. • The catalog page is not empty.
Trigger	User selects the movie from the movies catalog.
Minimal Guarantees	System ensures that the movie is selected.
Main Success Scenario	
User	1. Selects the movie from the movies catalog.
System	2. Displays all details of the selected movie (cover, title, genre, release year, director, cast, and ratings).
Exceptions	
2a - The system cannot show the movie details	
• 2a1. The system presents the message “No movie information.”	
Post-conditions	The user can see details of the selected movie.

Table 3: Use Case 003 - Categorize Movies

Use Case 003 - Categorize Movies	
Description	Allows the user to sort and filter movies by genre, author/director, and average rating.
Scope	Movies Catalog
Level	User Level
Preconditions	• User has access to the movies catalog page. • The catalog page is not empty.
Trigger	User opens the catalog page.
Minimal Guarantees	No movie is deleted.
Main Success Scenario	
User	1. Accesses the movie catalog.
System	2. Displays the list of movies with title, genre, release year, and cover.
System	3. Displays options for filtering and sorting.
User	4. Filters and sorts according to one or more criteria.
System	5. Displays the movies catalog according to the user’s criteria.
Exceptions	
2a - The system does not have the movies	
• 2a1. The system displays the message “No movies are available yet.”	
5a - There are no movies matching the selected criteria	
• 5a1. The system displays the message “No movies match these criteria.”	
Post-conditions	The user sees the movies according to the selected criteria.

Search Functionality

Table 4: Use Case 004 - Search Movies

Use Case 004 - Search Movies	
Description	Allows the user to search movies by using title, genre, and director, or both.
Scope	Movies Search
Level	User Level
Preconditions	User has access to the movies.
Trigger	User opens the search interface.
Minimal Guarantees	No movie is deleted.
Main Success Scenario	
User	1. Accesses the search interface.
User	2. Searches movies by using title, genre, and director, or both.
System	3. Retrieves the movies that match the specified search criteria from the database.
System	4. Displays up to 20 movies that match the specified search criteria.
User	5. Sees the movies that match the specified search criteria.
Exceptions	
3a - There are no movies matching the search criteria • 3a1. The system displays the message “No movies match these criteria.”	
Post-conditions	The user sees the movies according to the search criteria.

Table 5: Use Case 005 - Rate Movie

Use Case 005 - Rate Movie	
Description	Allows a registered user to rate a movie on a scale of 1–5 stars and optionally leave a comment.
Scope	Ratings Management
Level	User Level
Preconditions	• User is authenticated and logged in. • The selected movie exists in the catalog.
Trigger	• User selects a rating value for a specific movie.
Minimal Guarantees	• No existing data is lost if an error occurs during submission.
Main Success Scenario	
User	1. Navigates to the movie’s detail page.
System	2. Displays the current average rating, total number of ratings, and the user’s previous rating (if any).
User	3. Selects a score (1–5 stars) and optionally adds a comment.
System	4. Validates the input (rating range and user session).
System	5. Stores or updates the rating in the database.

System	6. Recalculates and updates the movie's average rating in real time.
System	7. Confirms the operation with the message: "Your rating has been saved successfully."
Exceptions	
4a - User not logged in 4a1. System displays: "Please log in to rate this movie." 4b - Invalid rating value (outside 1–5) 4b1. System displays: "Please select a valid rating between 1 and 5." 5a - Database connection or write error 5a1. System displays: "Unable to save your rating at this time. Please try again later." 6a - System fails to update the average rating 6a1. System displays: "Your rating was saved, but averages may take a few moments to update." 3a - User attempts to rate a movie that no longer exists 3a1. System displays: "Selected movie is unavailable."	
Post-conditions	The user's rating and comment are stored in the system, reflected in the movie's details, and included in future recommendation calculations.

Table 6: Use Case 006 - Browse Movies with Genre-Based Recommendations

Use Case 006 - Browse Movies with Genre-Based Recommendations	
Description	Allows the user to receive movie recommendations based on their preferred genres.
Scope	Recommendation Catalog
Level	User Level
Preconditions	The user has access to the home page with recommended movies.
Trigger	The user opens the home page.
Minimal Guarantees	No movie is deleted.
Main Success Scenario	
User	1. Accesses the home page.
System	2. Displays recommended movies based on genres that the user has rated 4–5 stars.
User	3. Navigates through the list of recommended movies.
Exceptions	
2a - User has not provided any ratings 2a1. System does not display genre-based movie recommendations.	
Post-conditions	The user sees movies recommended according to their favorite genres.

Table 7: Use Case 007 - Display Recommended Movies

Use Case 007 - Display Recommended Movies	
Description	Allows the user to view up to 10 recommended movies, each displayed with its title, genre, and average rating.
Scope	Movie Recommendation Display
Level	User Level
Preconditions	• The user has rated at least one movie, or the system can display top-rated movies for new users. • The recommendation list has been generated successfully.
Trigger	• User navigates to the “Recommended Movies” section or home page.
Minimal Guarantees	• The system displays available content even if some movie details are missing.
Main Success Scenario	
User	1. Opens the “Recommended Movies” page.
System	2. Retrieves up to 10 recommended movies from the database.
System	3. Displays each movie with its title, genre, and average rating.
User	4. Views the list and may select a movie to see more details.
Exceptions	
2a - No recommendations available • 2a1. System displays: “No recommendations available at this time.”	
2b - Database error • 2b1. System displays: “Unable to load recommendations. Please try again later.”	
Post-conditions	• The user views a list of recommended movies, each showing title, genre, and average rating.

AI-Generated Use Cases:

Prompt: For each Use Case presented below: <use cases>

- Improve the clarity
- Identify and add the missing exceptions
- Maintain the structure of the use case

Return only the use cases

Table 8: Use Case 001 - Browse Movies

Use Case 001 - Browse Movies	
Description	Allows the user to browse movies presented in the catalog.
Scope	Movies Catalog
Level	User Level
Preconditions	User has access to the movies catalog page.
Trigger	User opens the catalog page.
Minimal Guarantees	No movie is deleted.
Main Success Scenario	
User	1. Accesses the movie catalog.
System	2. Displays the list of movies with title, genre, release year, and cover.
User	3. Navigates through the list.
User	4. Selects the movie.
System	5. Shows a detailed page of the selected movie.
Exceptions	
2a - The system does not have the movies	
• 2a1. The system presents the message “No movies are available yet.”	
Post-conditions	The user can see the available movies.

Table 9: Use Case 002 - View Movie Details

Use Case 002 - View Movie Details	
Description	Allows the user to view details of a selected movie.
Scope	Movies Catalog
Level	User Level
Preconditions	• User has access to the movies catalog page. • The catalog page is not empty.
Trigger	• User selects a movie from the catalog.
Minimal Guarantees	• System ensures that the movie is selected.
Main Success Scenario	
User	1. Selects a movie from the catalog.
System	2. Displays all details of the selected movie, including cover, title, genre, release year, director, cast, and ratings.
Exceptions	

2a - System cannot show the movie details • 2a1. System displays the message “No movie information available.” 1a - User selects a movie that no longer exists • 1a1. System displays the message “Selected movie is not available.”	
Post-conditions	The user can see details of the selected movie.

Table 10: Use Case 003 - Categorize Movies

Use Case 003 - Categorize Movies	
Description	Allows the user to sort and filter movies by genre, director, and average rating.
Scope	Movies Catalog
Level	User Level
Preconditions	• User has access to the movies catalog page. • The catalog page is not empty.
Trigger	• User opens the catalog page.
Minimal Guarantees	• No movie is deleted.
Main Success Scenario	
User	1. Accesses the movie catalog.
System	2. Displays the list of movies with title, genre, release year, and cover.
System	3. Displays options for filtering and sorting.
User	4. Filters and sorts according to one or more criteria.
System	5. Displays movies according to the user’s criteria.
Exceptions	
2a - System does not have movies • 2a1. System displays the message “No movies are available yet.” 4a - No movies match the selected criteria • 4a1. System displays the message “No movies match these criteria.” 5a - System fails to update the catalog display • 5a1. System displays the message “Unable to display movies at this time.”	
Post-conditions	The user sees the movies according to the selected criteria.

Table 11: Use Case 004 - Search Movies

Use Case 004 - Search Movies	
Description	Allows the user to search for movies by title, genre, director, or a combination of these.
Scope	Movies Search
Level	User Level
Preconditions	• User has access to the movies.

Trigger	• User opens the search interface.
Minimal Guarantees	• No movie is deleted.
Main Success Scenario	
User	1. Accesses the search interface.
User	2. Searches movies using title, genre, director, or a combination.
System	3. Retrieves movies that match the search criteria from the database.
System	4. Displays up to 20 movies that match the search criteria.
User	5. Sees the movies that match the search criteria.
Exceptions	
3a - No movies match the search criteria • 3a1. System displays the message “No movies match these criteria.”	
3b - System cannot access the database • 3b1. System displays the message “Unable to retrieve movies at this time.”	
2a - User enters invalid or empty search criteria • 2a1. System prompts “Please enter valid search criteria.”	
Post-conditions	• The user sees movies according to the search criteria.

Table 12: Use Case 005 - Rate Movie

Use Case 005 - Rate Movie	
Description	Allows a registered user to rate a movie on a scale of 1–5 stars and optionally leave a comment.
Scope	Ratings Management
Level	User Level
Preconditions	• User is authenticated and logged in. • The selected movie exists in the catalog.
Trigger	• User selects a rating value for a specific movie.
Minimal Guarantees	• No existing data is lost if an error occurs during submission.
Main Success Scenario	
User	1. Navigates to the movie’s detail page.
System	2. Displays the current average rating, total number of ratings, and the user’s previous rating (if any).
User	3. Selects a score (1–5 stars) and optionally adds a comment.
System	4. Validates the input (rating range and user session).
System	5. Stores or updates the rating and comment in the database.
System	6. Recalculates and updates the movie’s average rating in real time.
System	7. Confirms the operation with the message: “Your rating has been saved successfully.”
Exceptions	

4a - User not logged in 4a1. System displays: “Please log in to rate this movie.”	
4b - Invalid rating value (outside 1–5) 4b1. System displays: “Please select a valid rating between 1 and 5.”	
5a - Database connection or write error 5a1. System displays: “Unable to save your rating at this time. Please try again later.”	
6a - System fails to update the average rating 6a1. System displays: “Your rating was saved, but averages may take a few moments to update.”	
3a - User attempts to rate a movie that no longer exists 3a1. System displays: “Selected movie is unavailable.”	
Post-conditions	The user’s rating and comment are stored in the system, reflected in the movie’s details, and included in future calculations.

Table 13: Use Case 006 - Browse Movies with Genre-Based Recommendations

Use Case 006 - Browse Movies with Genre-Based Recommendations	
Description	Allows the user to receive personalized movie recommendations based on their preferred genres.
Scope	Recommendation Catalog
Level	User Level
Preconditions	- The user has access to the home page with recommended movies. - The user has previously rated movies (optional, affects recommendations).
Trigger	The user opens the home page.
Minimal Guarantees	- No movie is deleted from the catalog. - The system remains operational even if recommendations cannot be generated.
Main Success Scenario	
User	1. Accesses the home page.
System	2. Displays recommended movies based on genres that the user has rated highly (4–5 stars).
User	3. Navigates through the list of recommended movies.
Exceptions	
2a - User has not provided any ratings 2a1. System displays a generic list of popular or trending movies instead of personalized recommendations.	

2b - Recommendation engine is unavailable • 2b1. System informs the user that personalized recommendations are temporarily unavailable. • 2b2. System may display a fallback list of popular or trending movies.	
Post-conditions	• The user sees a list of movies either personalized according to their favorite genres or as a fallback popular list. • The system logs any recommendation failures for maintenance purposes.

Table 14: Use Case 007 - Display Recommended Movies

Use Case 007 - Display Recommended Movies	
Description	Allows the user to view up to 10 AI-generated movie recommendations based on the user's rating history, preferences, and behavior patterns. Each movie is shown with its title, genre, and average rating.
Scope	Personalized Movie Recommendations
Level	User Level
Preconditions	• The user is logged in. • The AI recommendation engine has access to user ratings and genre preferences.
Trigger	• User navigates to the "Recommended for You" section or home dashboard.
Minimal Guarantees	• The system displays top-rated movies as a fallback if personalized data is unavailable.
Main Success Scenario	
User	1. Opens the "Recommended for You" section.
System	2. Analyzes the user's previous ratings and viewing history.
System	3. Selects up to 10 movies with similar genres or high predicted user interest.
System	4. Displays each movie with its title, genre, average rating, and a note such as "Because you liked Inception."
User	5. Views the list and can select a movie for more details or add it to their watchlist.
Exceptions	
2a - No user rating data available • 2a1. System displays top 10 highest-rated movies (minimum of three total ratings).	
3a - AI service unavailable • 3a1. System displays: "Recommendations currently unavailable. Showing popular movies instead."	
2b - Database access error • 2b1. System displays: "Unable to load recommendations at this time."	

Post-conditions	• The user views personalized movie recommendations. • The AI engine may update its model based on user interactions.
------------------------	--

3.4 Non-Functional Requirements

Traditional

Performance

NFR1: The system should respond to user requests (e.g., login, search, recommendation) within 2 seconds under normal load.

Scalability

NFR2: The system should handle at least 20 concurrent users without performance degradation.

Availability

NFR3: The system should be available at least 95% of the time during demonstration.

Security

NFR4: User passwords must be stored in encrypted form.

NFR5: Only authenticated users can access personalized features (profile, recommendations, rating).

Usability

NFR6: A new user should be able to register, log in, and find a movie within 5 minutes without external help.

Maintainability

NFR7: The codebase should follow consistent structure and naming conventions.

Portability

NFR8: The system should run in any environment that supports Docker.

Reliability

NFR9: The system should recover gracefully from invalid input without crashing.

AI-Assisted

Context was given. Prompt: I have already defined some non functional requirements. Review these requirements and flag any that are ambiguous, subjective, or not measurable and rewrite them. Suggest additional requirements. Give me a final list.

Performance

NFR1: The system must respond to user requests (login, search, recommendation) within 2 seconds when handling up to 20 concurrent users with an average database response time of 500ms.

Scalability

NFR2: The system must support at least 20 concurrent users without response time exceeding 3 seconds or error rate exceeding 1%.

Availability

NFR3: The system must maintain 95% uptime during the demonstration period, with planned downtime communicated at least 24 hours in advance.

Security

NFR4: User passwords must be hashed using bcrypt (or an equivalent algorithm with a minimum work factor of 10) before storage.

NFR5: All personalized features (profile, recommendations, ratings) must require valid session authentication; unauthenticated requests must return HTTP 401.

NFR6: User sessions must expire after 24 hours of inactivity or when the user logs out.

Usability

NFR7: A new user must be able to complete registration, login, and find a movie within 5 minutes, measured through user testing with a success rate of at least 80%.

NFR8: The system must be accessible via modern browsers (Chrome, Firefox, Safari, Edge — latest two versions).

Maintainability

NFR9: Code must pass automated linting checks (as defined in the project's linter configuration) with zero errors before deployment.

NFR10: All API endpoints must include basic documentation (purpose, parameters, and response format) either in code comments or in separate API documentation.

Portability

NFR11: The system must run on any platform supporting Docker Engine version 20.10 or higher and Docker Compose version 2.0 or higher.

Reliability

NFR12: The system must handle invalid input (malformed requests, invalid data types) by returning appropriate error messages (4xx status codes) without crashing or exposing stack traces.

NFR13: Database transactions for critical operations (user registration, ratings) must be atomic — either completing fully or rolling back completely.

Monitoring

NFR14: The system must log all authentication attempts, errors, and critical operations with timestamps and user context (where applicable).

4 System Design

4.1 C4 System Context Diagram

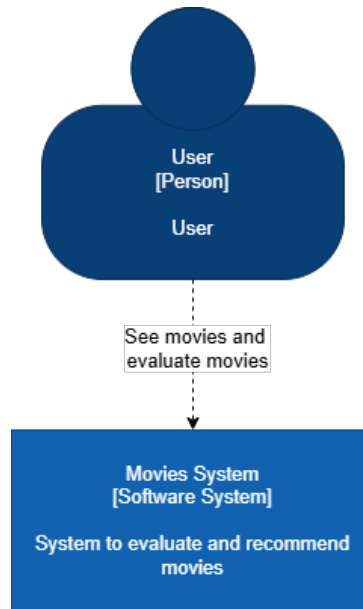


Figure 1: System Context Diagram

- Actors: user, admin, system.
- Use cases: register, login, browse, search, rate, recommend, manage catalog.

4.2 Architecture Overview

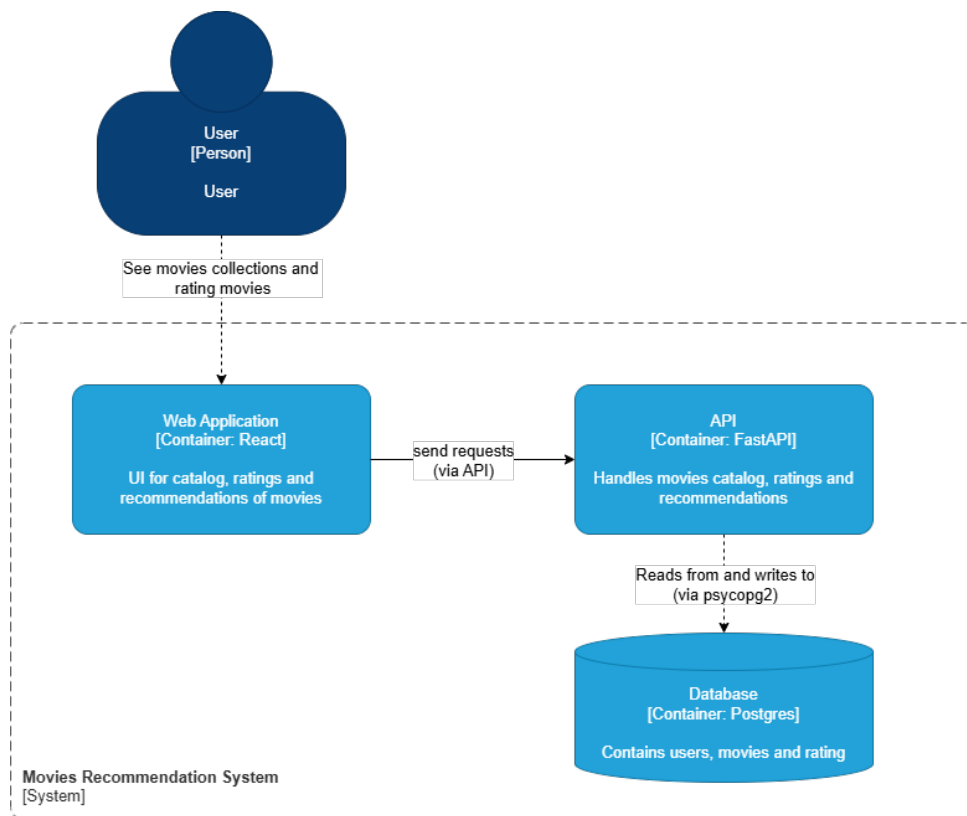


Figure 2: Architecture Overview

4.3 Database Design (Conceptual)

Traditional:

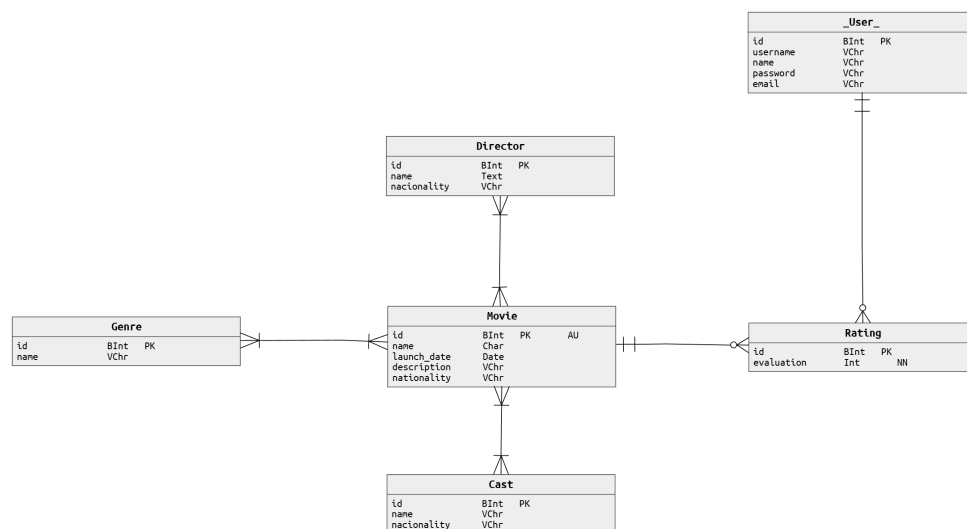


Figure 3: Traditional Entity Relationship

AI-Assisted:

Prompt: Conceptual diagram: <generated conceptual diagram json>

Based on the provided ER Conceptual Diagram, enhance it:

- suggest new fields for these features:
- determine Not Null attributes
- ensures that links are corrected

Return only the JSON of the ER diagram (not representing the FK or schemas related to links)

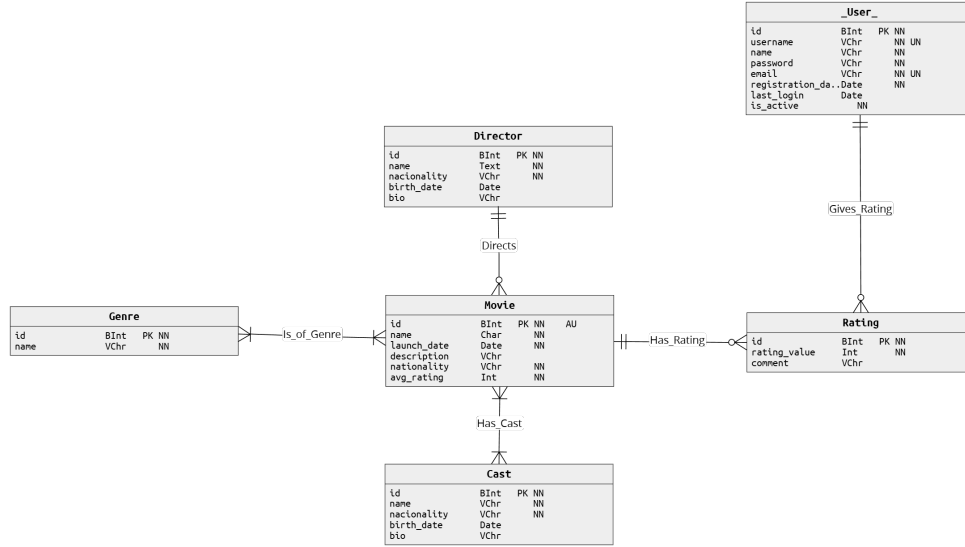


Figure 4: AI-Assisted Entity Relationship

The LLM added the new fields, for instance, the birth date and bio for each Cast and Director. In the case of Rating, LLM added a comment field to allow users to add a comment while rating a movie. For optimisation of queries, LLM added the avg_rating on the movie schema that should be updated when any user rates the movie.

5 Assumptions and Constraints

5.1 Assumptions

1. User Environment
 - Users will have a stable internet connection while accessing the system.
 - Users will access the system primarily through modern web browsers (Chrome, Firefox, Safari, Edge) that support JavaScript and cookies.
2. Data & Catalog
 - The movie catalog data will be sourced from a predefined dataset or trusted APIs (TMDb).
 - Movie metadata (genre, release year, cast, etc.) is assumed to be accurate and complete from the source.
 - Ratings are assumed to reflect user preferences honestly (no deliberate spam/bot ratings initially).
3. Users & Usage
 - The system will initially support a small to medium user base (100 concurrent users) before scaling.
 - New users without prior ratings will receive non-personalized recommendations (e.g., trending/popular movies).
 - Admins will be available to manage catalog integrity (removing duplicates, fixing incorrect entries).
4. Recommendation Engine
 - Personalized recommendations are assumed to improve over time as user data grows.
 - Initial recommendation algorithms may be rule-based or collaborative filtering.
 - Feedback loops (likes/dislikes, updated ratings) will refine the recommendation quality progressively.

Additional Assumptions (Generated by LLM)

- Users are expected to create accounts using valid emails; disposable or temporary emails may be restricted.
- Recommendation quality depends on the volume of collected user ratings; accuracy will be limited at early adoption.
- The system will not initially support offline access; all functionality requires an active connection to the server.
- The database size is assumed not to exceed 500,000 movies in the first release.

5.2 Constraints

1. Technical Constraints

- The system must be deployed in a containerized environment (e.g., Docker).
- Database technology will be limited to SQL (e.g., PostgreSQL/MySQL) for structured storage of users, ratings, and movies.
- APIs must conform to RESTful standards and be documented with OpenAPI/Swagger.
- System must integrate with CI/CD pipelines for automated deployment.

2. Performance Constraints

- Search and recommendation results must be delivered within 2 seconds under normal load.
- The recommendation engine must update within 1 minute after new ratings are added.
- This scenarios will be shown on the quality attributes scenarios.

3. Security & Privacy Constraints

- User passwords must be hashed and salted (b64 encryption).
- Only authenticated users can access personalized features (profile, recommendations, ratings).
- The system must comply with data protection guidelines (e.g., GDPR).
- Sensitive operations (login, rating submission) must use HTTPS.

4. Operational Constraints

- System availability must be 95% uptime during demo and 99% in production.
- Backup and recovery mechanisms must exist to restore data in case of system failure.
- Initial deployment limited to English, but system must support multi-language extension in future iterations.
- The recommendation model must be explainable to some degree (e.g., "Recommended because you liked X").

5. Resource Constraints

- The development team is small, with limited availability for implementing advanced features.
- Only open-source or internally licensed software and tools will be used.
- Only free or open data sources (e.g., TMDb free tier) can be integrated for catalog information.
- Testing resources are limited, and only essential functionality will be fully validated in the first release.
- CI/CD pipelines will be implemented using self-hosted tools with limited automation capabilities, restricting advanced deployment features such as auto-scaling or multi-environment rollouts.

Additional Constraints (Generated by LLM)

- Recommendation models must provide results within the hardware constraints of a single on-premise server (no distributed computing).
- Error handling will focus on preventing system crashes; graceful degradation of features (e.g., fallback to trending movies if recommendations fail) must be supported.
- Logging and monitoring will be restricted to critical system events due to limited storage and processing resources.
- No integration with external payment or subscription systems is planned in the MVP.

Differences written by us and generated by LLM

- The original assumptions and constraints (written by us) focused mainly on basic user expectations, catalog accuracy, recommendation improvement, technical deployment choices, and small-team limitations.
- The additional assumptions and constraints (generated by LLM) expand the scope by adding practical operational limits (e.g., database size, no offline access), early-stage limitations (recommendations weak until enough data exists), and fallback strategies (default to trending if recommendations fail). They also highlight future scalability challenges and restrictions on integrations (e.g., no payments, no external SaaS).